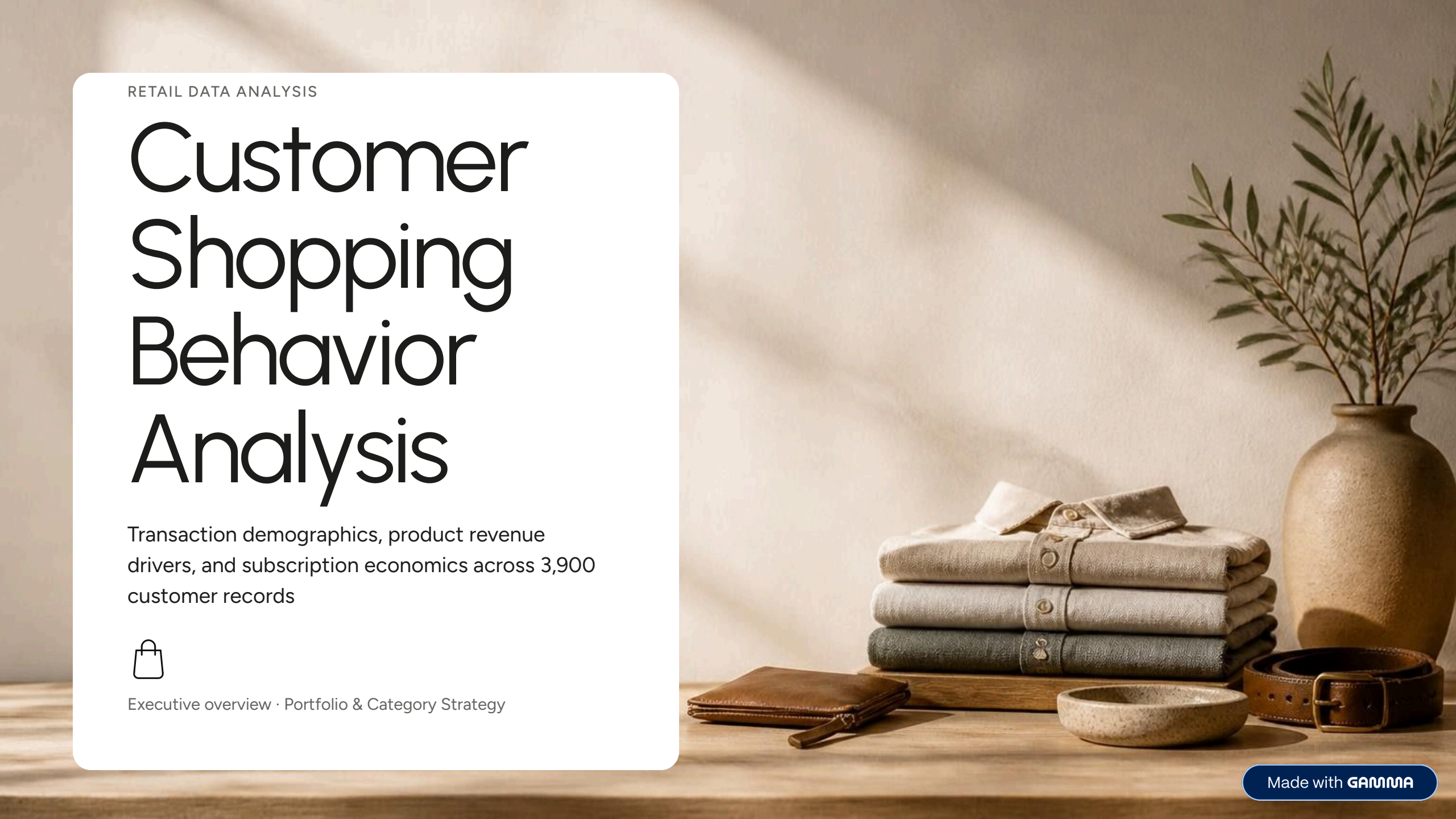
RETAIL DATA ANALYSIS

Customer Shopping Behavior Analysis

Transaction demographics, product revenue drivers, and subscription economics across 3,900 customer records

Executive overview · Portfolio & Category Strategy



Made with GAMMA

EXECUTIVE SUMMARY

3,900 purchases reveal who spends and why

A transactional study examining consumer spending patterns, customer segments, category preferences, and subscription retention dynamics.

CORE OBJECTIVE

Uncover spending patterns, high-value segments, and subscription dynamics

TRANSACTION SCOPE

3,900 retail transactions spanning four distinct product categories

Analyzed across demographic, product, and historical purchase dimensions.



Made with GAMMA

DATA ARCHITECTURE

The dataset: 3,900 rows, 18 columns

Complete transaction logs covering demographic, item-level, and historical behavioral variables

DIMENSION 01

Demographics

Customer age, gender, geographic location, and subscription status capturing the core profile.

DIMENSION 02

Purchase details

Item purchased, category classification, purchase amount, season, apparel size, and color variant.

DIMENSION 03

Behavior & Ratings

Discount usage, previous order count, purchase frequency cadence, review ratings, and shipping type.

Data completeness: High fidelity across all fields; the only observed gap was 37 missing review ratings, addressed during data cleaning.



18 structured attributes providing comprehensive transaction and customer tracking.

Made with GAMMA

DATA PIPELINE

From raw data to a clean database

Five-stage automated pipeline converting raw transactional flat files into a relational schema

01

Profiling

Loaded raw transactional file and profiled distributions using pandas.

02

Imputation

Filled 37 missing review ratings using each product category's median.

03

Normalization

Standardized all 18 column names to uniform snake case.

04

Engineering

Engineered age group cohorts and discrete purchase frequency features.

05

Storage

Dropped redundant promo code column, loaded into PostgreSQL

Schema verified and indexed in PostgreSQL for downstream business queries

Production-grade ETL ensuring consistency across analytics and reporting pipelines.

Made with GAMMA

KEY PERFORMANCE METRICS

Headline numbers: $59.76 average purchase, 3.75 rating

01.

3.9K

Total customers

Individual buyers represented across transaction logs

02.

$59.76

Average purchase

Consistent order size across diverse catalog items

03.

3.75

Average review rating

5-point scale satisfaction metric across all orders

04.

27%

Subscribers

27% enrolled in membership; 73% non-subscribers

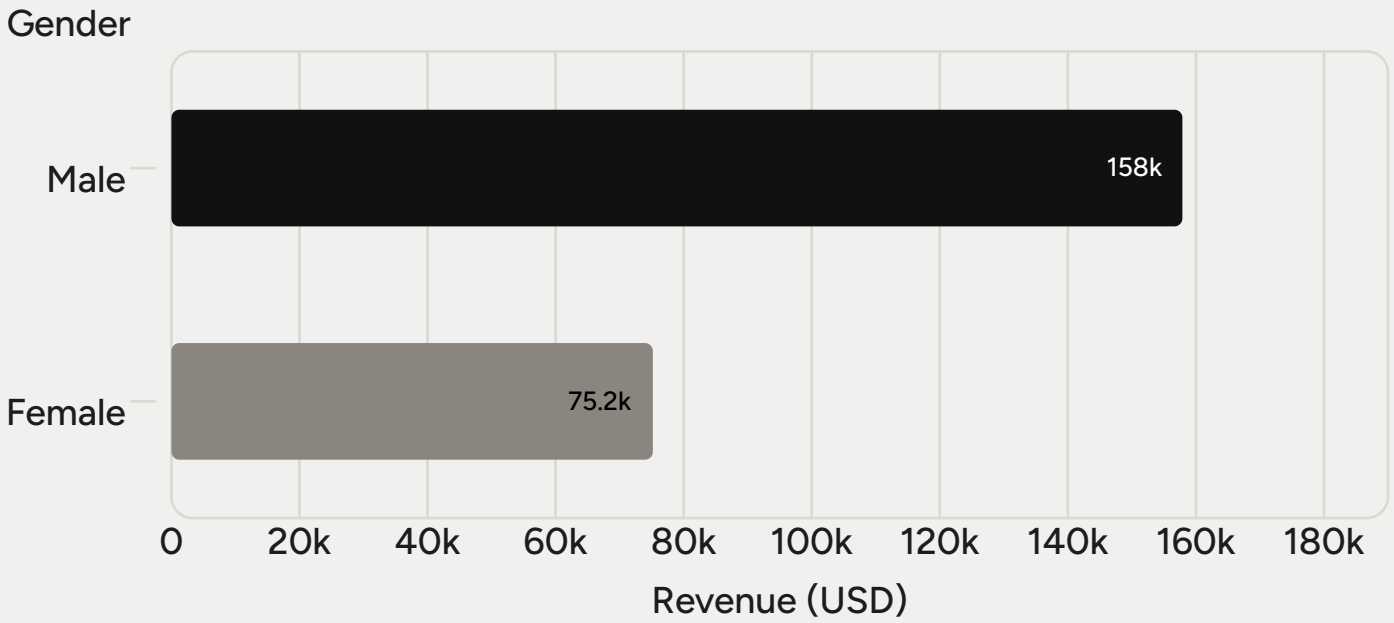


27% subscribers; 73% non-subscribers.

Made with GAMMA

DEMOGRAPHIC BREAKDOWN

Male customers generate about twice the revenue of female customers



Total gross revenue split by customer gender (USD)

KEY TAKEAWAY

2.1x revenue multiple

Male shoppers contributed **$157,890** compared to **$75,191** from female shoppers, driving over 67% of total company gross sales.

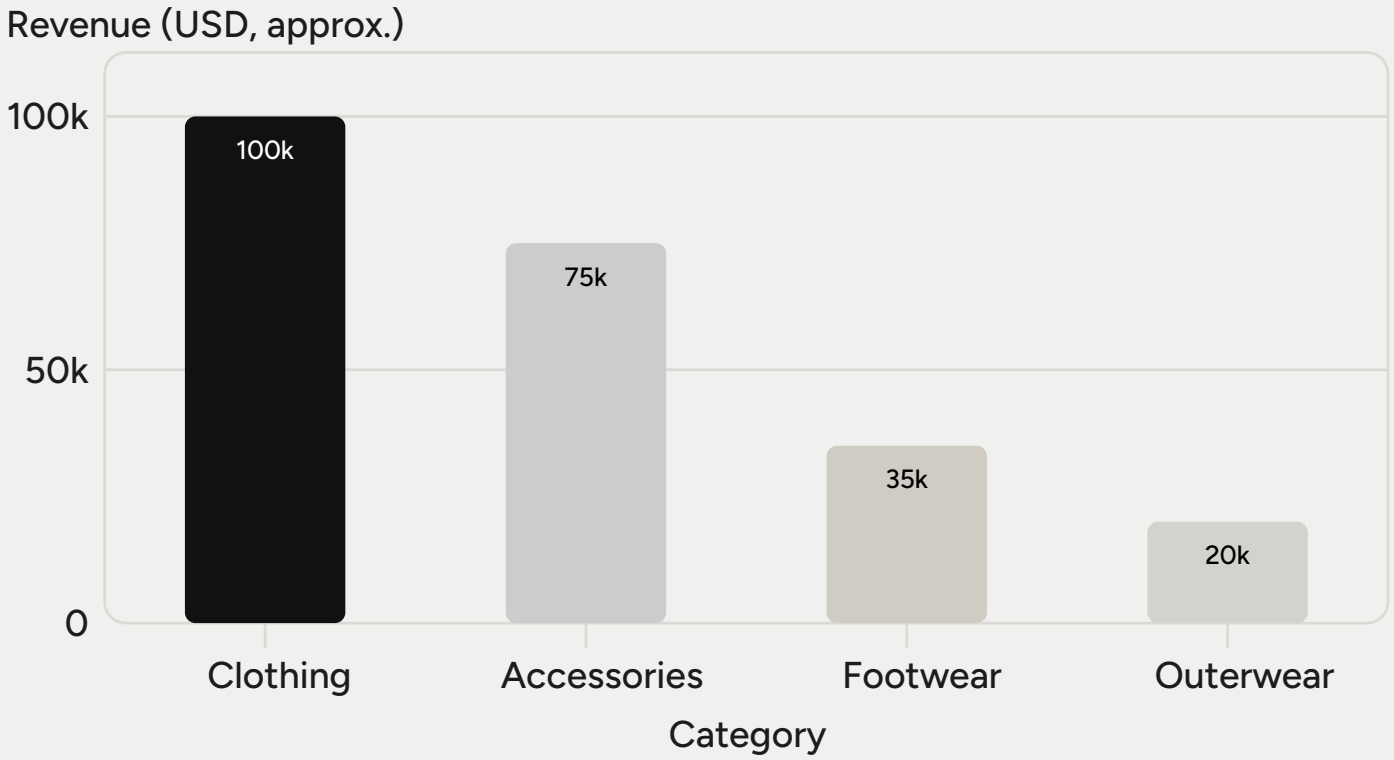
Transaction counts and buyer volumes in the sample reflect a predominantly male active purchasing base.

Marketing acquisition shows significant upside potential in balancing female customer acquisition.

Made with GAMMA

CATEGORY PERFORMANCE

Clothing and accessories drive the revenue



Gross merchandise revenue by product classification (USD, approximate)

VOLUME & SALES CORRELATION

Clothing leads; outerwear trails

Top two categories account for over **75% of sales**. Clothing (~$100K) and accessories (~$75K) vastly outperform footwear (~$35K) and outerwear (~$20K).

Sales order volume follows the identical hierarchy, showing demand concentration in staple daily wear.

Sales volume follows the same order.

Made with GAMMA

PRODUCT COMPARISON

Best sellers and top-rated products differ

Order volume peaks in essentials and jewelry, while customer satisfaction scores peak in seasonal accessories

Most ordered item	Top rated item
Jewelry (171)	Gloves (3.86)
Blouse (171)	Sandals (3.84)
Pants (171)	Boots (3.82)
Jacket (163)	Hat (3.80)
Sandals (160)	Skirt (3.78)

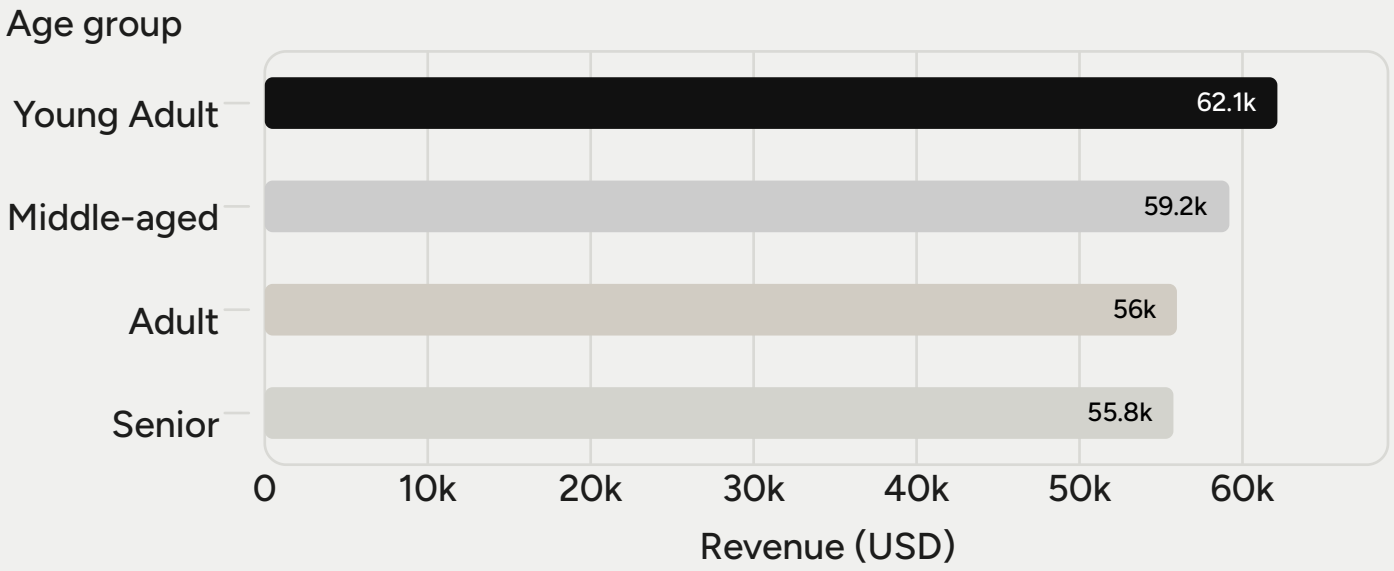
Strategic note: Only sandals appear on both top lists (160 orders, 3.84 rating). Highest-rated items like gloves and boots show strong customer loyalty despite moderate volume.

Divergence highlights opportunity to cross-promote top-rated niche items alongside volume drivers.

Made with GAMMA

AGE COHORT ANALYSIS

Young adults lead revenue, but age groups sit close



Aggregate revenue generated across demographic age cohorts (USD)

SPREAD EVALUATION

Narrow 10% spread across cohorts

Young adult $62,143; middle-aged $59,197; Adult $55,978; senior $55,763.

Made with GAMMA

SUBSCRIPTION BEHAVIOR

Subscribers spend the same as everyone else

Basket sizes show zero subscription premium, but repeat purchaser retention rates remain solid

Subscribers

1,053 customer members (27%)

AVERAGE TICKET

$59.49 per purchase

TOTAL REVENUE

$62,645 gross revenue

REPEAT SHOPPERS

958 repeat buyers subscribe

Non-subscribers

2,847 customer members (73%)

AVERAGE TICKET

$59.87 per purchase

TOTAL REVENUE

$170,436 gross revenue

REPEAT SHOPPERS

2,518 repeat buyers do not

Difference in average spend is just $0.38 · Program currently drives membership, not larger baskets

Opportunity to introduce tiered member perks or threshold incentives to expand subscriber order size.

Made with GAMMA